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| **Fiscal Year:** | FY 2021 | **Task Last Updated:** | FY 09/03/2020 |
| **PI Name:** | Rosenblatt, Charles S. Ph.D. | | |
| **Project Title:** | Microgravity Studies of Nanoparticles and Topological Defects in Liquid Crystal Thin Films | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | COMPLEX FLUIDS/SOFT MATTER--Complex Fluids | | |
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| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station |
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| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | Liquid Crystal Facility | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Lacaze, Emmanuelle Ph.D. ( Université Pierre et Marie Curie, France )<br>Hegman, Torsten Ph.D. ( Kent State University )<br>Taylor, Philip Ph.D. ( Case Western Reserve University )<br>Lemieux, Robert Ph.D. ( COLLABORATOR / University of Waterloo ) | | |
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| **Task Description:** | Nanoparticle / liquid crystal composites have created new avenues for scientific discovery and technological development. For example, recent studies have shown that mixtures of liquid crystals and appropriate semiconducting nanoparticles can be exploited for potential high efficiency and easily manipulatable photovoltaic devices. But the development of this and other technologies requires a fundamental understanding of how the nanoparticles incorporate and move within the liquid crystal, how they aggregate in and around topological defects, and how the nanoparticles affect the defects themselves. We have shown that fluorescent carbon dots (FCDs) may be dispersed in liquid crystals and used as a probe of defects, demonstrating that they accumulate in regions of high liquid crystal curvature so as to relieve the local strain. As a result their emitted light can serve as a quantitative indicator for the interaction and evolution of topological defects. Interpretation of these initial results, however, is compromised by the presence of substrates, which is required in an Earth-bound environment. Here we will perform experiments in the absence of a container by utilizing the Observation and Analysis of Smectic Islands In Space (OASIS) module aboard the International Space Station (ISS). By studying thin flat liquid crystal films doped with FCDs -- this will be done by dissolving the FCDs in a solvent and "shooting" slowly moving pico droplets of the mixture at the liquid crystal film -- in a microgravity environment, we will obviate the need for bounding substrates, thereby rendering a much simpler system for interpretation. In particular, we will use OASIS' cameras to study the spatial distribution and transport of the FCDs in and around topological defects in the smectic-C phase of the liquid crystal, especially close to the smectic-A phase transition temperature. A related project will be undertaken in which the film will be bombarded by droplets of methanol solvent from a pico inkjet printer involving a large mechanical impulse. The impact of the droplets will induce the formation of "islands" and "divots" of smectic liquid crystal. The goal of this part of the project will be to determine the size and thickness of these induced structures as a function of the film thickness and the droplet speed. The work will involve both experimental and theoretical components. The results will provide a clear picture of how nanoparticles distribute themselves over time, how defects in the liquid crystal interact, and how islands form on a liquid crystal film. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Liquid crystal (LC) textures and topological defects have been of interest for well over a century. Defects in two-dimensions generally are categorized in terms of the "defect strength"-- "s". Studies of defects can be used to obtain values of LC anchoring strength coefficients and elastic constants, albeit with limited accuracy. Defects are central in establishing the director profile in geometrically-confined liquid crystals, and serve as an important signature when determining the symmetry of phases, for example, distinguishing between a biaxial nematic phase and a Smectic-C (Sm-C) phase. Defect dynamics provide another important field of study, as defect motion is extremely sensitive to boundary effects and provides information about surfaces and impurities. Island and divot formation are useful in understanding the layer structure of the smectics and how layers reorganize. They provide information about the surface tension of the liquid crystal.<br>Perhaps most important use of topological defects is that they can serve as traps for micro- and nanoparticles (NPs), as the presence of the particles reduces or eliminates the divergent energy associated with the defect core. In other words, nanoparticles (NPs) migrate within the LC in the direction of increasing stress associated with the liquid crystal's orientational order, as the presence of the particles provides relief from the large stresses that would be present at LC singularities. The incorporation of NPs into liquid crystals and their subsequent behavior in and around defects finds applications in areas such as energy harvesting, photonic structures, mirrorless lasing, micro-patterned LC devices, and chiral metamaterials for photonics and communications applications. They also may have applications in pharmaceuticals, as an alternative means of drug delivery. |
| **Task Progress:** | The worldwide emergence of the SARS-CoV-2 virus and resulting Covid-19 had a deleterious effect on our – and indeed everybody's – effort, although we attempted to move forward with mostly calculational work, along with limited laboratory work.<br>We have been investigating film thickness measurement techniques, and have focused on an approach involving illumination with a quasi-white light source (but having a known spectral distribution) and imaging of the reflected light using the R, G, and B channels of a CCD (charge-coupled device) camera, also of known spectral response. We have modeled the intensity vs. film thickness for each of the channels, and developed a calculational technique to extract the film thickness and avoid the ambiguity of an intensity that is periodic in film thickness by using multiple color sensors in the detector. Moreover, this approach is superior to the laser illumination approach as it allows us to image an area, and extract the local film thickness when islands, divots, or steps in thickness are present. We also have written software to account for ancillary effects, such as the angular spread of the light and noise. An automated image processing macro was developed to extract intensity values from film images. This work was conducted partially at Case and partially remotely during the summer of 2020, with undergraduate Rachel Margulies working with Dr. Padetha Tin at NASA (as well as with the Principal Investigator-PI Charles Rosenblatt at Case).<br><br>Currently we are investigating the possibility of using one or multiple LEDs (light emitting diodes) for illumination, which would obviate the need to account for the illumination spectrum of the quasi-white light. Reflection intensity measurements are being conducted on films of various thickness that contain steps and islands.<br><br>We have continued some of the work from the previous year in order to finalize the materials to be used aboard ISS. We have been adjusting the voltage parameters to our inkjet head to optimize the droplet size and speed for methanol and IPA (isopropyl alcohol). This has resulted in a decrease in the size of the islands created, which is essential for our fluorescent carbon dot experiment to study topological defects in smectic-C films using the liquid crystals 9OO4 and ZA-284. But importantly, the control of the ejected methanol droplets from the inkjet printer allows us to study the creation of islands and divots in the film. (The solvent evaporates quickly and is chosen to have poor solubility with the liquid crystal. Thus, the island formation is due to the impulse force of the droplet.) In particular, the diameter and thickness of the islands and divots are being examined as functions of the film's original thickness and the droplet speed. We have performed 1g studies of island and divot formation, and have concluded that we will use only normal incidence for the experiments aboard the ISS. Moreover, we have decided to perform the island/divot experiments using the liquid crystal ZA-284 and methanol.<br><br>Using Dr. Tin's film holder, which for us has been an improvement over Prof. Yokoyama's design last year, and which relies on two plates with holes having razor-sharp edges passing over each other, we have been drawing films at various spreader speeds in order to assess empirically how the approximate film thickness and stability is related to the speed. We have found that there is little correlation between spreading speed and film thickness, suggesting that the actual flight experiments will need to be performed by creating a film of arbitrary thickness, measuring the thickness, |

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| | performing the experiment, breaking the film, and then creating another film of arbitrary thickness.<br><br>We examined electric field induced transition from a strength m = +1 partially escaped radial topological defect to a pair of split (m = ½) defects, and the same transition in the reversed direction. We found the transitions to be discontinuous (first order), and developed a theory for the phenomenon. In the course of the experiments we discovered the phenomenon of co-rotation of the split defects about a common point. This work is continuing. |
| **Bibliography Type:** | Description: (Last Updated: 12/04/2024) |
| **Articles in Peer-reviewed Journals** | Susser AL , Harkai S , Kralj S , Rosenblatt C. "Transition from escaped to decomposed nematic defects, and vice versa." Soft Matter. 2020 May 28;16(20):4814-22. https://doi.org/10.1039/d0sm00218f ; PMID: 32409816 , May-2020 |
| **Articles in Peer-reviewed Journals** | Murray BS, Harkai S, Rosenblatt C, Kralj S. "Electric field driven reconfigurable multistable topological defect patterns." Phys Rev Research. 2020 Feb-Apr;2(1):013176. https://doi.org/10.1103/PhysRevResearch.2.013176 , Feb-2020 |
| **Articles in Peer-reviewed Journals** | Ferris AJ, Afghah S, Selinger RLB, Selinger JV, Rosenblatt C. "Electric field-induced crossover from 3D to 2D topological defects in a nematic liquid crystal: Experimental verification." Soft Matter. 2020 Jan 21;16(3):642-50. https://doi.org/10.1039/C9SM01733J ; PMID: 31693053; PMCID: PMC6981022 , Jan-2020 |